Vision-enhanced Indoor Space Localization

Context Awareness & Location Sensing

- **Context Awareness**[1]: Computers can both sense, and react based on their environment.
- To Build Intelligent Systems – WHO WHAT WHEN WHERE and HOW.[2]

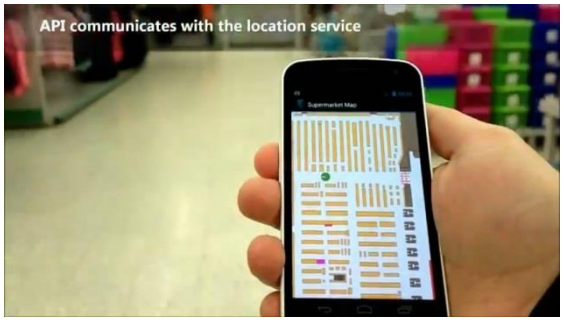
Category[3]	
Computing Context	*network connectivity, communication costs & bandwidth, nearby resources*
User Context	*user's profile, location, people nearby, even the current social situation*
Physical Context	*lighting, noise levels, traffic conditions, and temperature.*

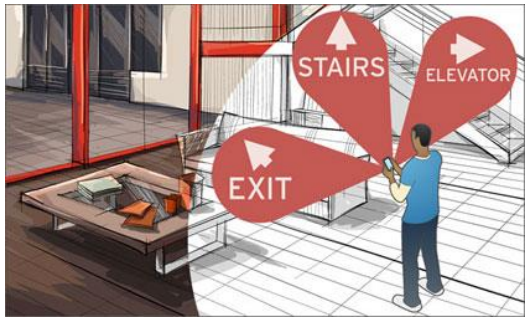
Location Sensing plays a fundamental role in multiple Context Awareness applications!!!

Indoor Localization

- Most recent studies to monitor location with smart phones make use of the embedded GPS and accelerators. (widely used in outdoors)[4]

- Location information obtained through GPS is only reliable in certain outdoor environments with direct visibility to the satellites.

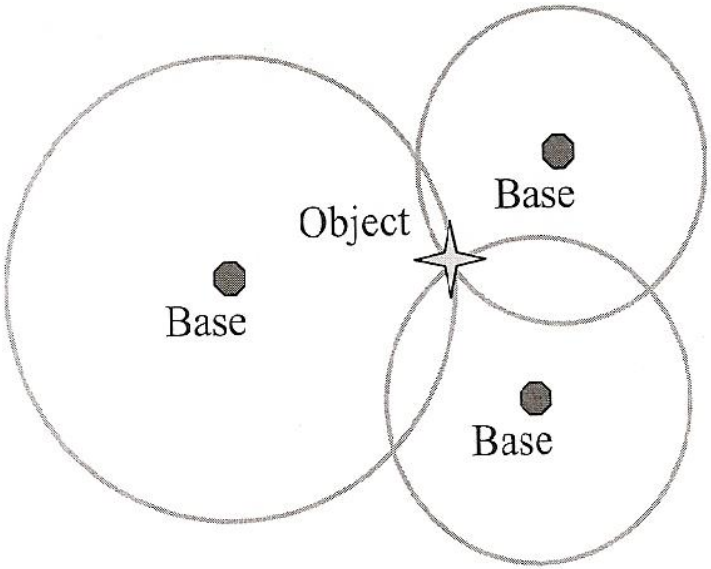




- Indoor localization technologies provide a user's location to deliver context-aware functionality, have become very popular in recent years.[5]

Radiofrequency Based Localization

- **RFID & Bluetooth: Signals can stay stabile and tags can be distinguished with each other.**
- **Convert the Signal levels to distance estimates according to a simple propagation model.**
- **Drawback: expensive to construct the infrastructures.**

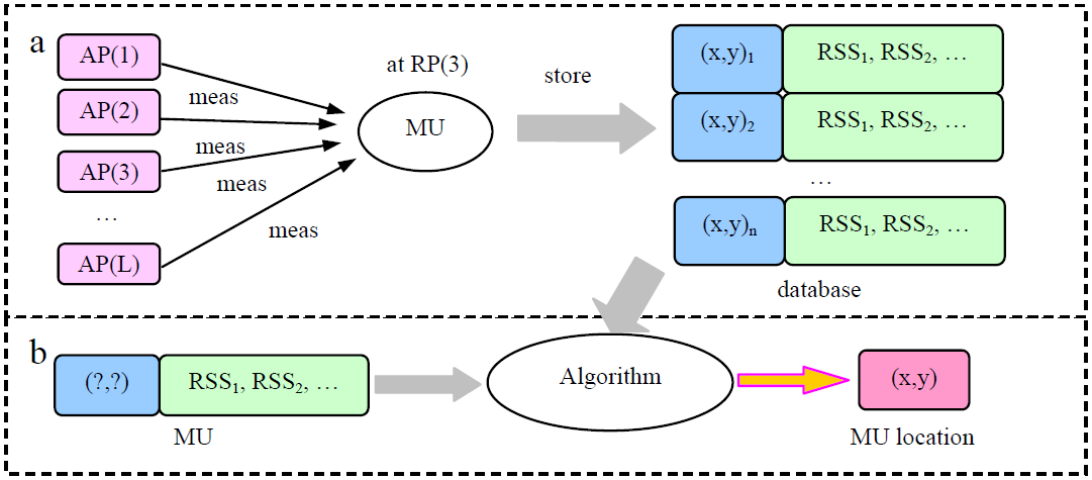


RFID tags used in LANDMARC prototype System

iBeacon developed by Apple.Inc.

WIFI Based Localization[7]

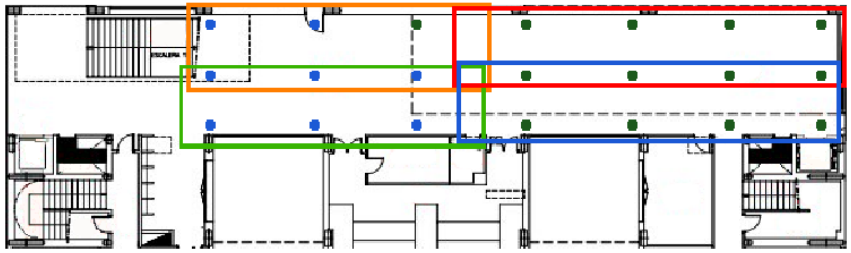
- growing rapidly as the flexibility, connectivity, mobility, and low cost meets the needs of consumers.
- Radio Maps: Mapping location-dependent parameters of measured radio signals in the area of interest.

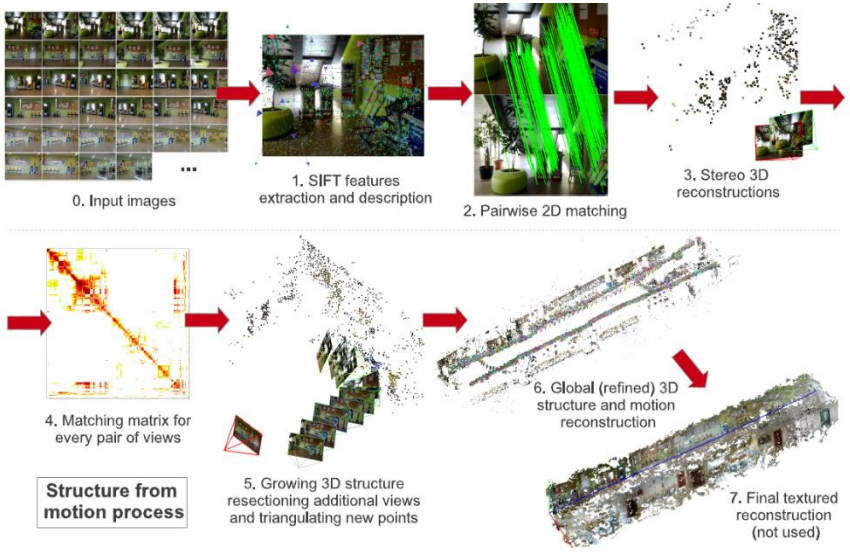


- **Drawback**: time-consuming and elaborate to collect WIFI signal strengths for creation of radio maps.

Camera Based Localization[8]

- Requires large training databases, the main challenge of this approach is to achieve real-time capability.
- Hybrid Approach[9][10]: WIFI is used to pruning the training database, Computer Vision Algorithm is used to infer a fine-grained location.



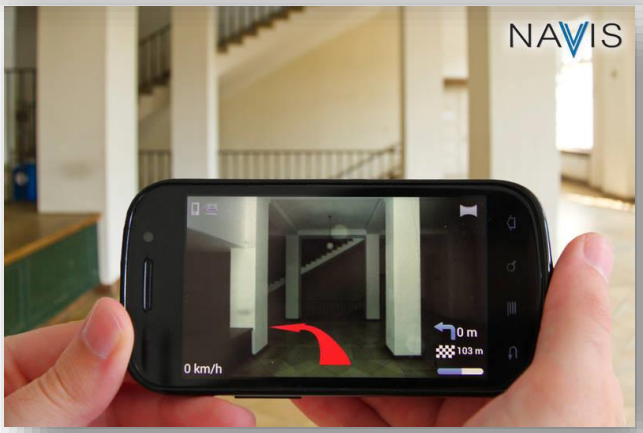


- Drawback: generating SIFT-based 3D models is time-consuming and unsuitable for mobile localization systems.

Challenges of Existing Techniques

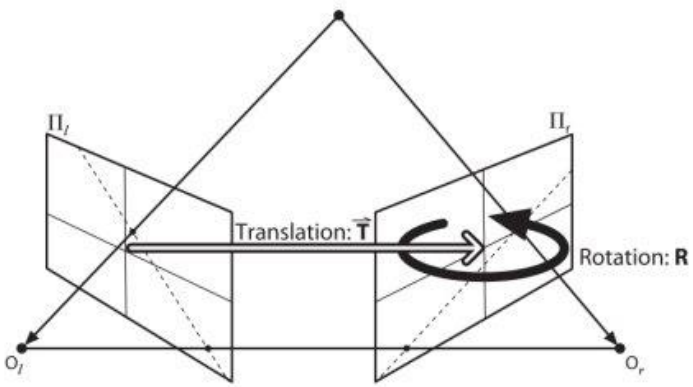
- **High expense** and manual labors to build the localization infrastructures.

- Not be able to be **intended for** indoor Augmented Reality Applications.



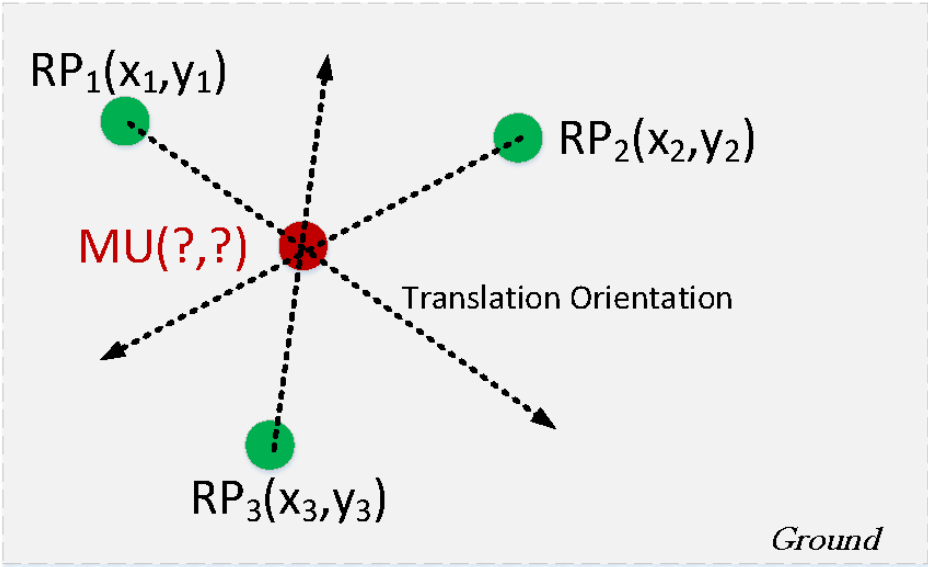
- Complicated offline configuration that demands **elaborate sampling** and **high-cost computation**.

Our Proposed Approach

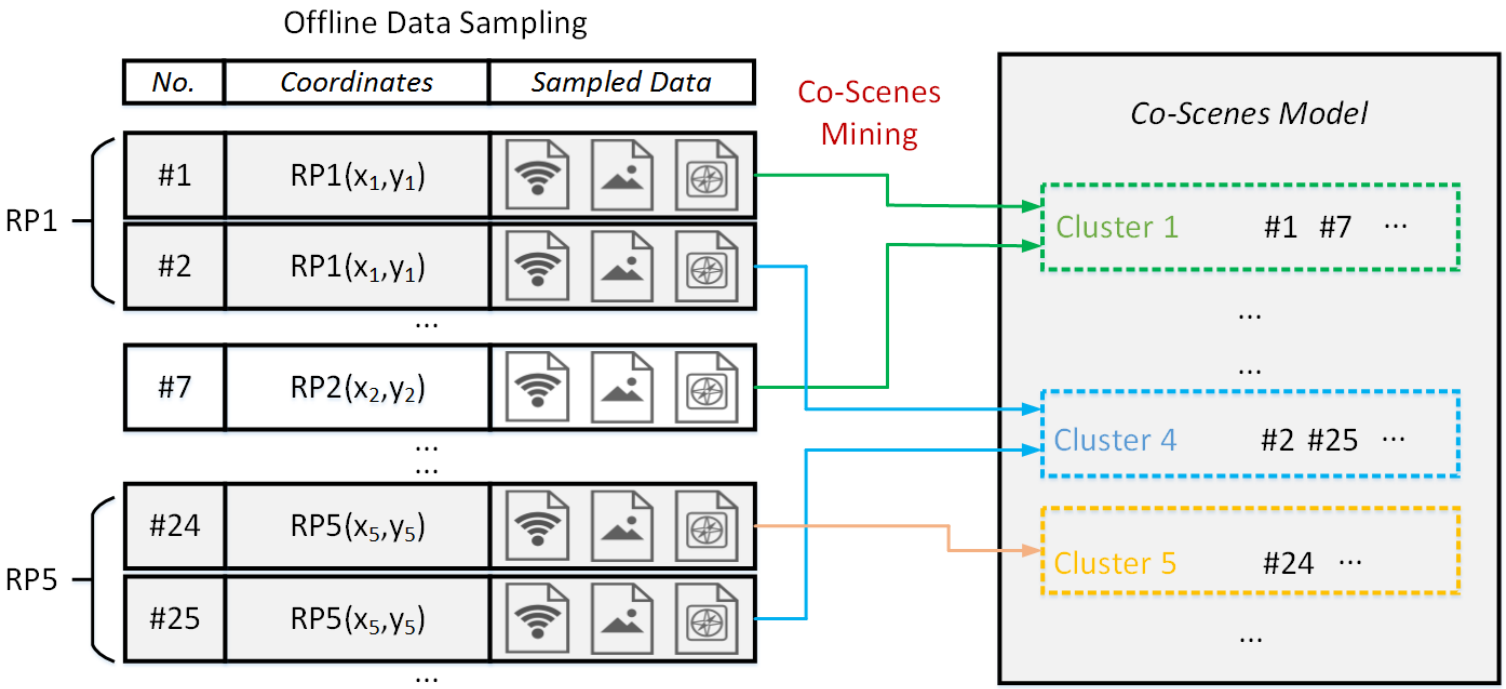


- **Epipolar Geometry**[11] implies the **spatial relationship** between two photos which are **overlapped in visual content**.
- Low-cost: ≥ 8 matching points pairs are needed.

- If the location of one RP is known(**green dot**), the query point(**red dot**) can be inferred with the estimated translation.



Offline Sampling Phase



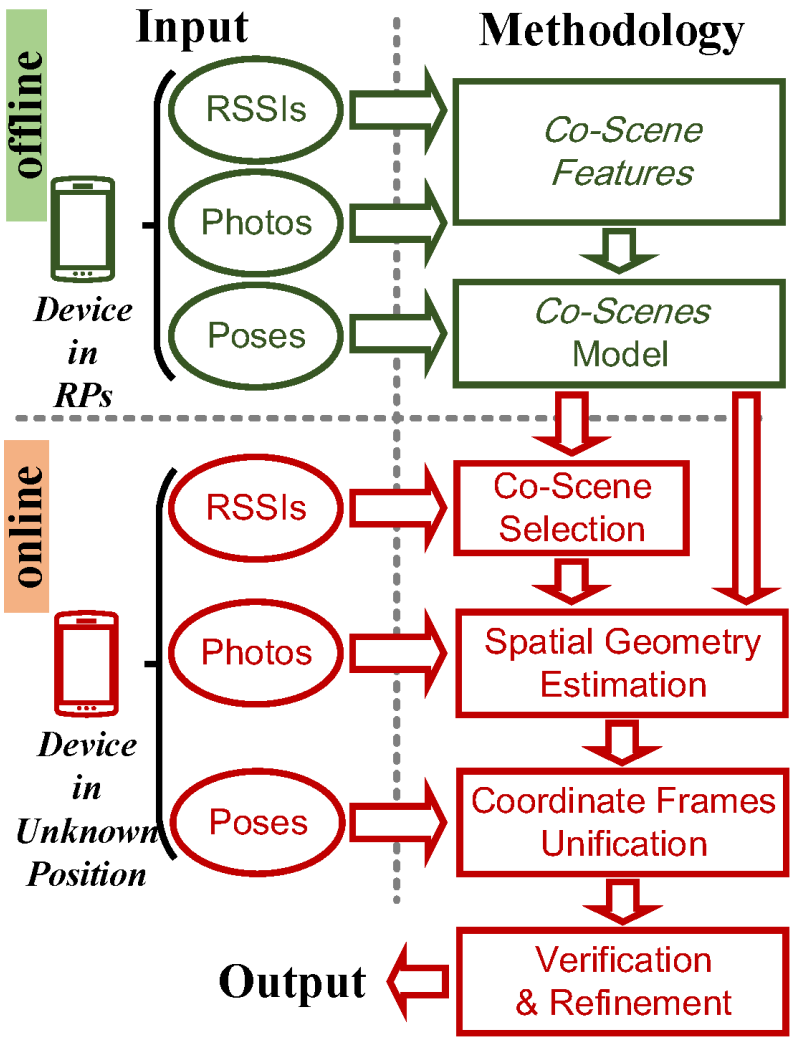
Sample each Reference Point by recording:
1) a few phone-captured photos;
2) corresponding camera poses;
3) A series of WIFI signals.

- **Co-Scenes:** a set of feature vectors which are neighbors in both visual and signal spaces.

- **Co-Scenes Mining**: Co-Scenes are clustered by computing the similarities between each feature vectors. The overall similarity is measured by a combination of similarities in both visual and WIFI spaces.

Online Query Phase

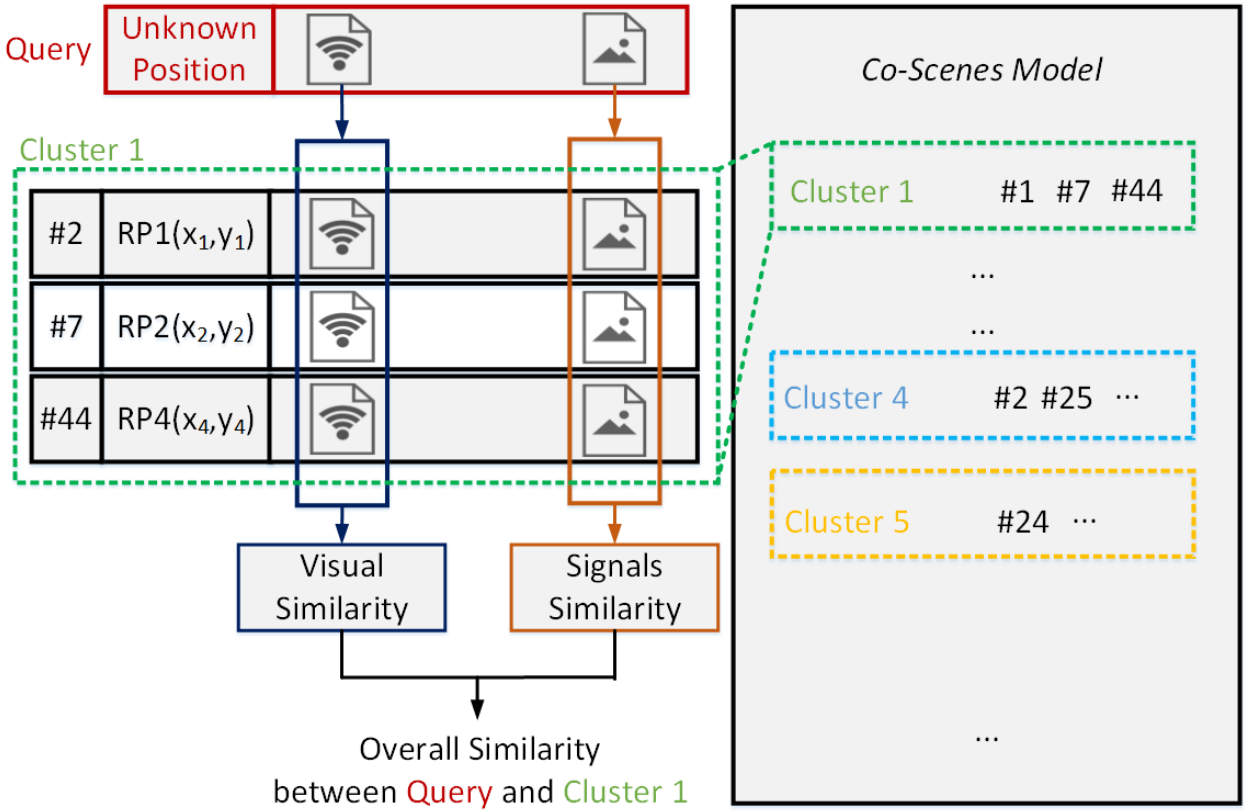
- Four steps are performed in the online query phase to infer a Mobile Unit's position.
- Step 1: **Co-Scene Selection** picks up the most similar Co-Scene of query.
- Step 2: Orientations between RPs and MU are obtained by epipolar based **Spatial Geometry Estimation**.
- Step 3: Obtained orientations are transformed into the Indoor Coordinate Frame.
- Step 4: **Verification & Refinement** step verifies the well-fitted orientations and obtain the refined ultimate result.



Co-Scene Selection

- When a query is launched with a phone-captured photo and a series of WiFi signals, we compute the similarities in both visual and WiFi spaces for each Co-Scene Cluster in the *Co-Scenes Model*.

- The overall similarity between a query and a Co-Scene is measured by a combination of above two similarities. As a result, the most similar Co-Scene is selected for the next step.



Spatial Geometry Estimation(SGE)

- **Fundamental Matrix *F*** describes the **spatial displacement** between two Co-Scene photos.

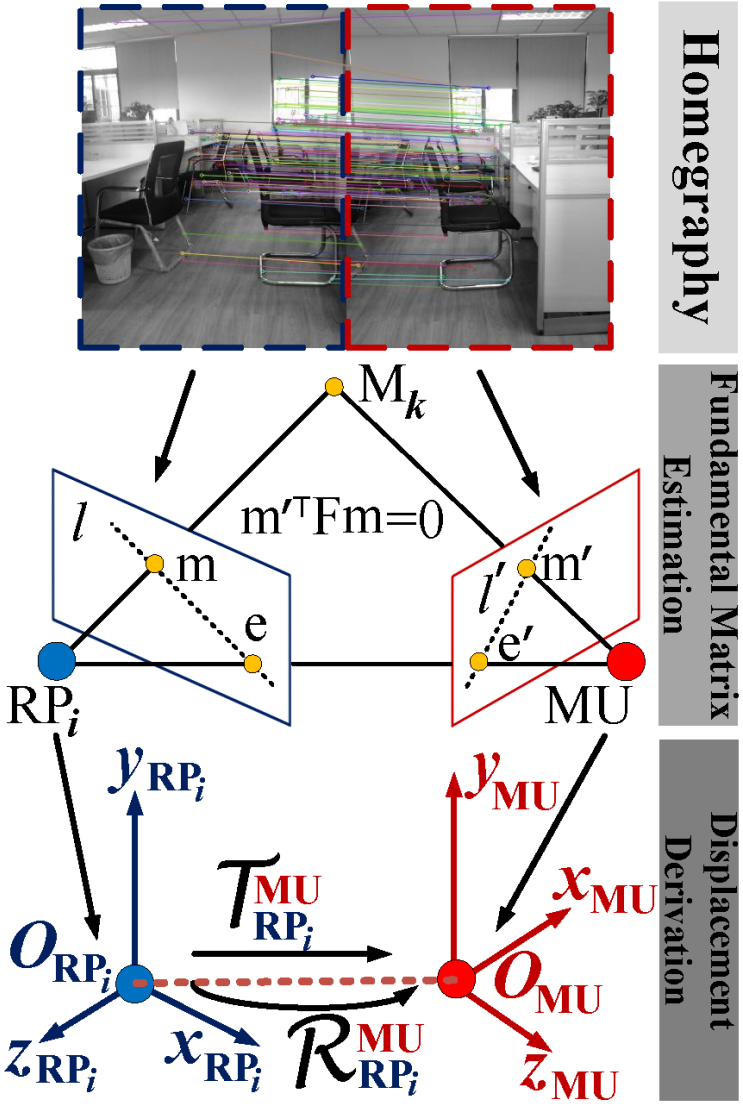
- Given sufficiently (≥8) matching points pairs m′↔ m from two photos, ***F*** can be estimated by the equation:

$$m'^T F m = 0$$

- ***F*** can be derived as:

$$F = K'^{-T}[\mathcal{T}]_\times \mathcal{R} K$$

- translation ***T*** and rotation ***R*** from the known location RP to unknown location MU can be obtained.



Outlier Rejection in Visual Matching

- **Outliers: The matches of points in two photos may be negative(red line in the left), the negative matches are considered as outliers in visual matching.**

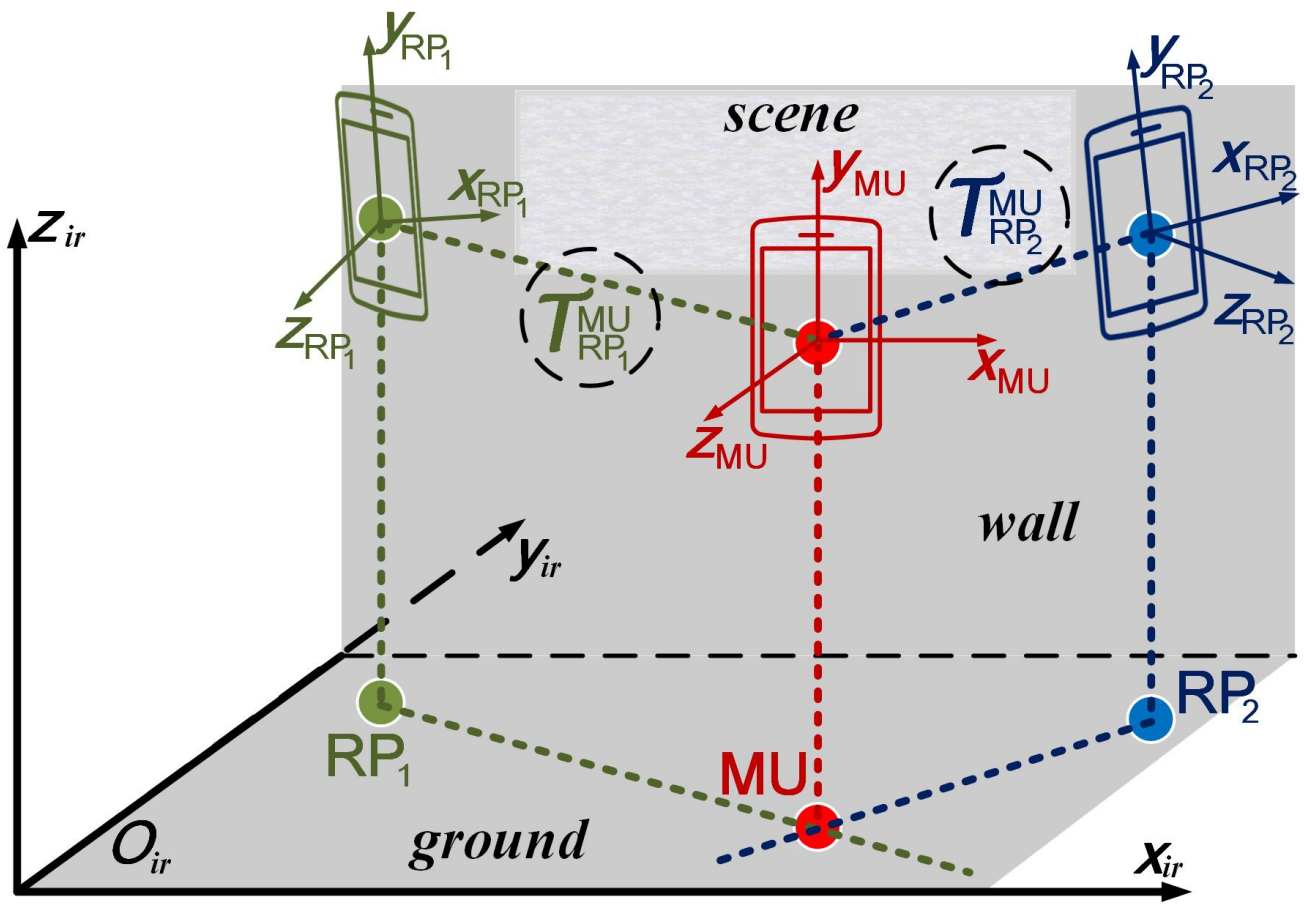
- **Outlier Rejection: Remove bad visual matching points pairs to get a much better estimate of Fundamental Matrix and further more the spatial relationship of two cameras.**

SIFT matching without Outlier Rejection

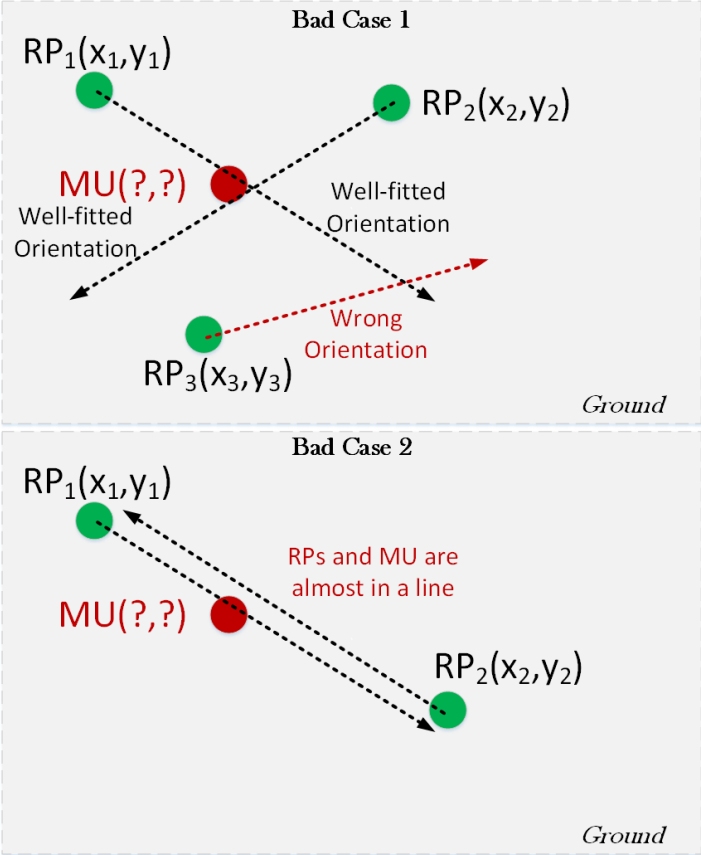
With Outlier Rejection, bad matching can be reduced

Triangulate the unknown position

- Query is launched by the Mobile Unit, RP1 and RP2 are selected as they refer to the most similar Co-Scene of MU.
- The translation relationship (orientation) between each RP and MU can be estimated with the captured photos when standing on those points.
- ≥ 2 orientations are needed since the real-world distance can not be obtained.



Verification & Refinement

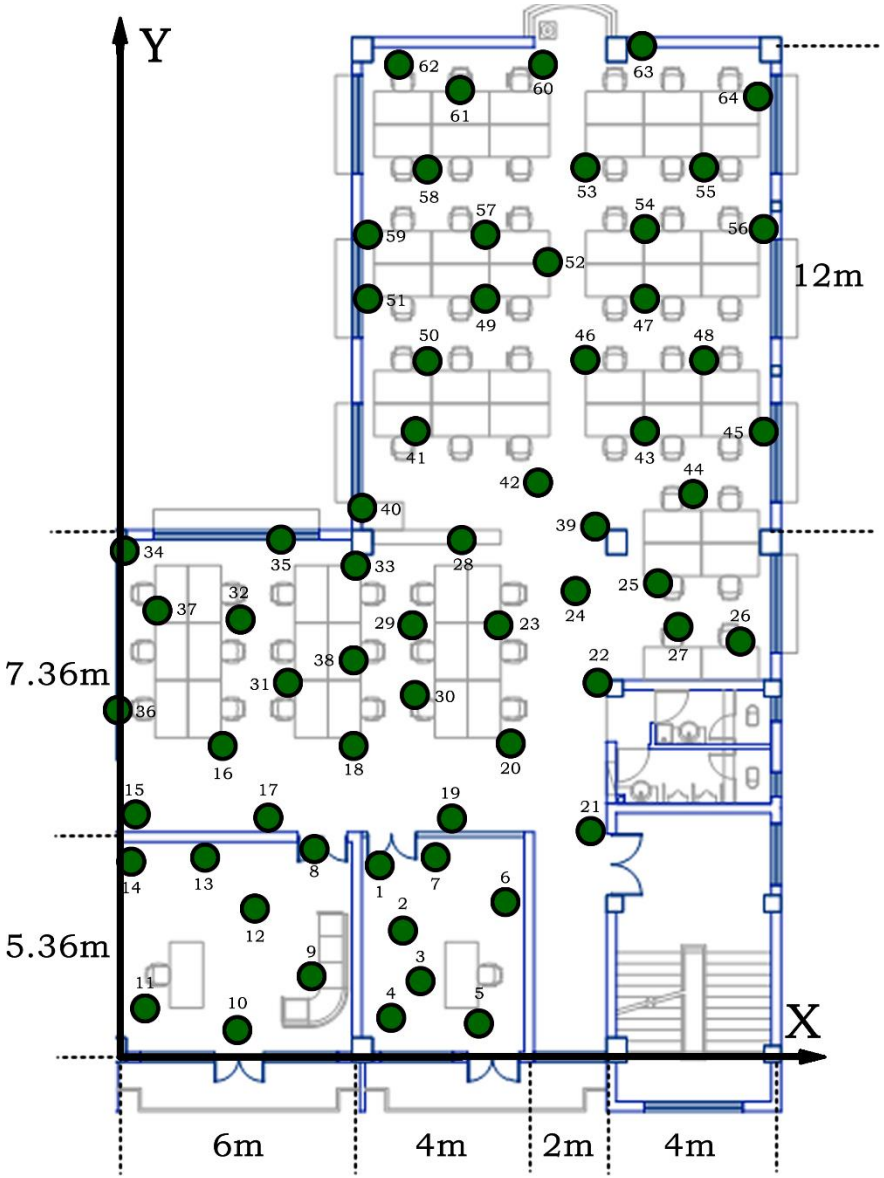


- Errors can be **accumulated** in computation of orientations ***T***, due to the **mismatching of images** (in SGE) and **noisy measurement** of gyroscopes (in CFU).
- Verification and Refinement procedure is needed to select **well-fitted orientation**.
- Since ***T*** and rotational displacement ***R*** are derived **simultaneously**, we can verify ***R*** **instead of** ***T*** .

Take advantage of Gyroscopes

```
R′ = Dot(Rotation Matrix(RP1->MU) , Gyro Rotation Matrix(RP1))
If(similarity(R′, Gyro Rotation Matrix(MU)) < threshold)
        Orientation(RP1->MU) is good orientation
```

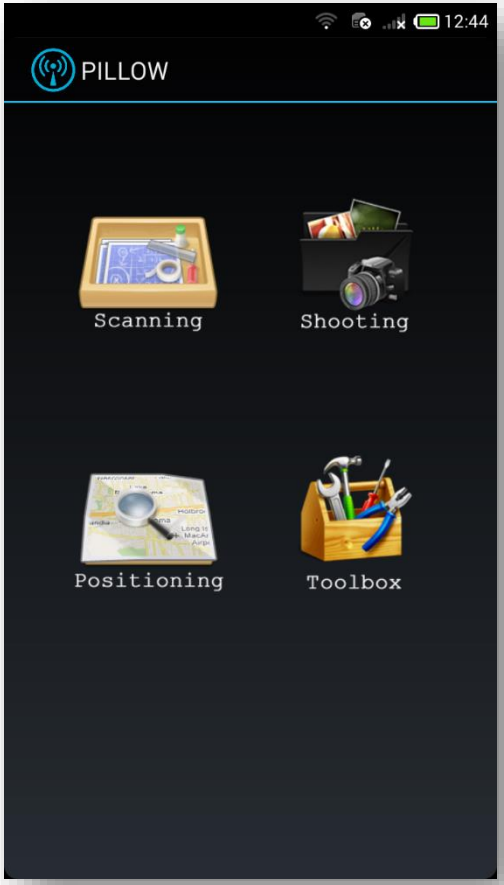
Test bed



- The floor-plan of 5th Floor, Tumu Building is provided.
- The dimension of the test bed is around 12m * 24.72m
- 64 RPs are selected, the coordinates of each RP are measured manually.
- The Wireless Environment is composed of 16 Access Points around.

Sampling Tool and Data Collection

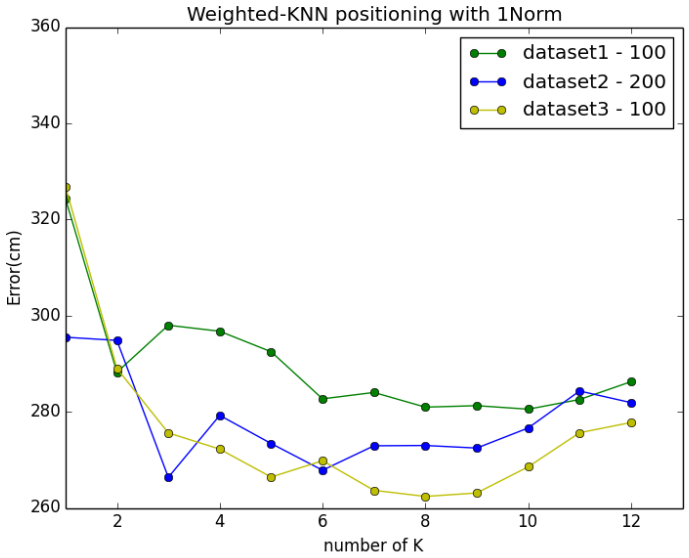
- We developed a sampling tool on an Android Smartphone to support our offline modeling.
- WIFI Signals Collection: Only 100 or 200 records for each RP, 20mins would be enough for the whole test bed (64 RPs).
- Photos Collection: The original image and its corresponding rotation matrix are recorded and uploaded to the server. 6 photos are captured in average for each RP.
- Photos are resized and SIFT features are extracted for all the resized photos.

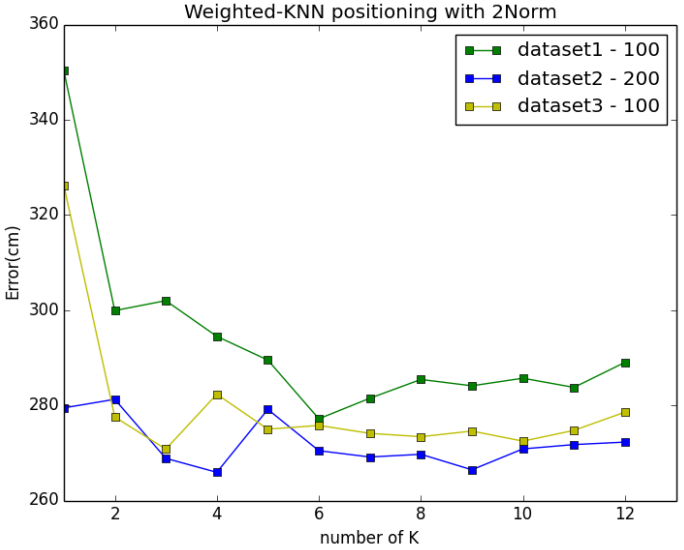


Localization with only WIFI signals

- **Weighted-KNN algorithm was implemented as the contrast method. The unknown position of the Mobile Unit can be inferred with only the collected WIFI signals.**

- **K nearest neighbors of the RPs are determined, the weight of each RP is calculated according to the similarity distance.**

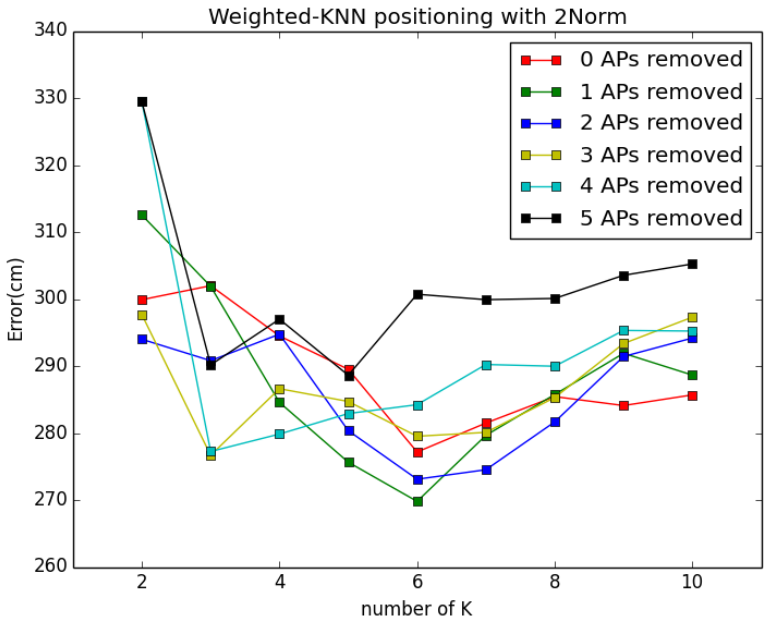




- **Three datasets are tested, The average localization accuracy of WKNN is ≥ 2.6m. The best can be achieved with parameter K = 6.**

APs Reduction

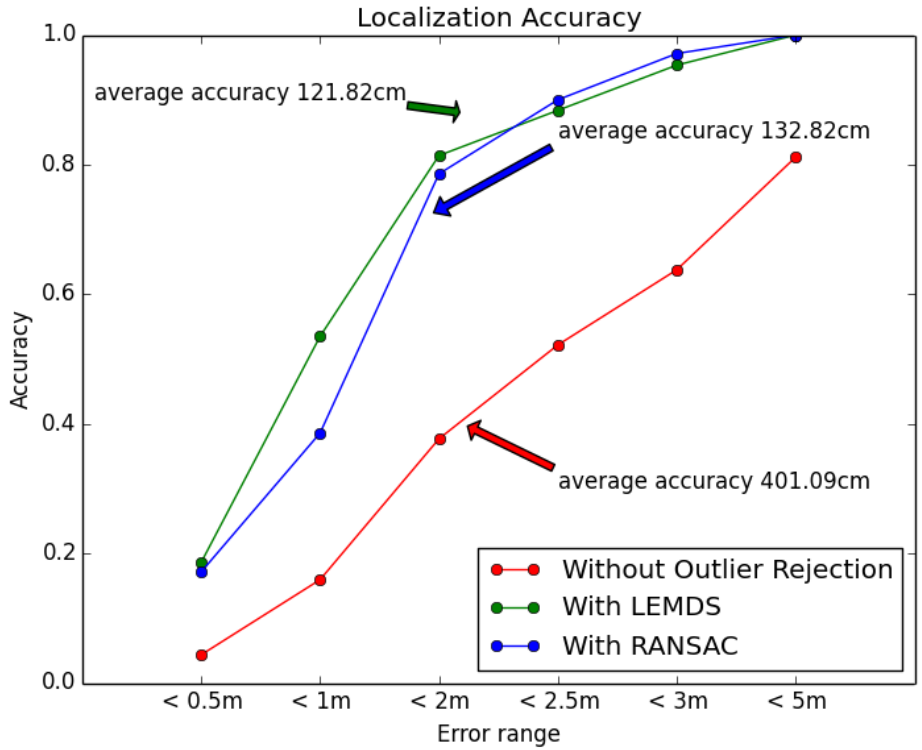
- A high number of APs may deteriorate accuracy because of the redundancy. SelectPercentile Algorithm is implemented in order to reduce the redundant APs.

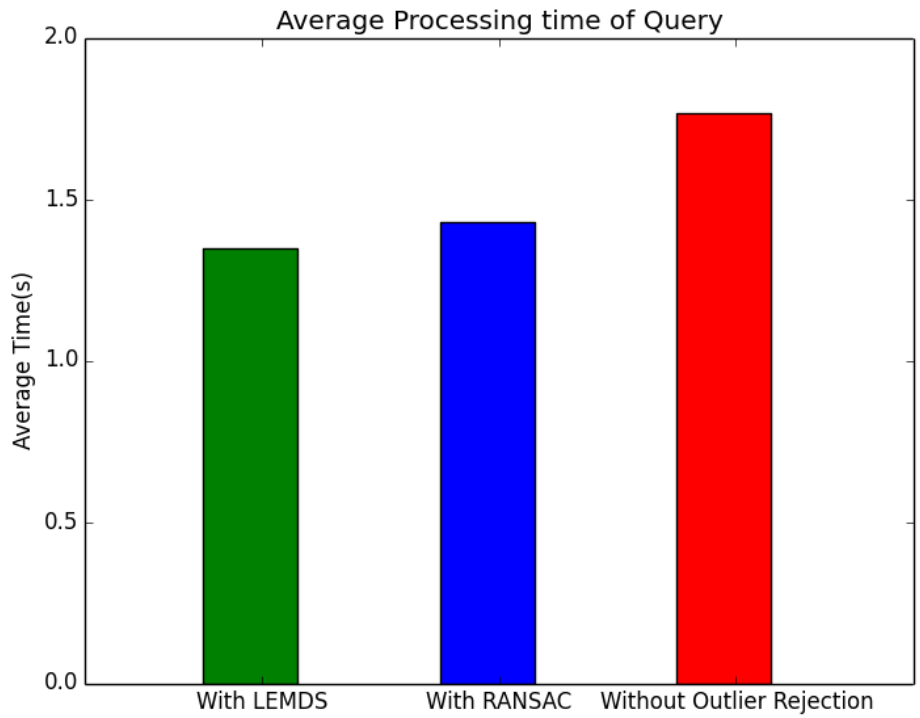


- Better localization accuracy can be achieved if certain APs are reduced. For example, When parameter K = 6, moving 1 or 2 APs will outperform the one without APs Reduction.

Effectiveness of Outlier Rejection

- **Two outlier rejection algorithms RANSAC and LEMDS(Least Median of Squares)**

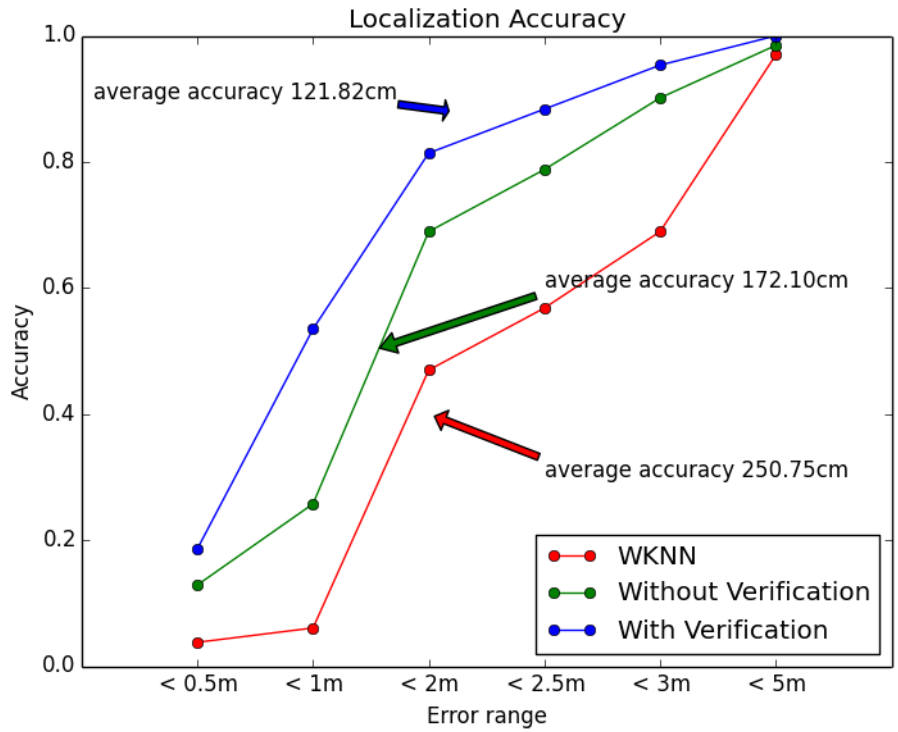




- **Without Outlier Rejection, the estimates of Epiploar Geometry are not good enough to be used in the triangulation. The localization accuracy can be significantly improved if Outlier Rejection is conducted after visual matching.**

Localization Accuracy & Precision

- **Cross Validation: A Reference Point and its sampled data(signals & photos) is selected as the query, the others are used to build the Co-Scenes Model.**



- **Our proposed method outperformed the WKNN algorithm with only WIFI signals. In our method, if verification operation is conducted, the average accuracy can be improved from 1.72m to 1.21m.**

Conclusions

- We introduced **Co-Scene**, a set of feature vectors which are neighbors in both visual and signal spaces.

- **Epipolar Geometry** implies the spatial relationship between two Co-Scene photos, providing a lightweight way to estimate accurate position at **significantly lower cost** in both **computation** and **manual labor**.

- The implemented prototype completes an entire cycle, from sensing to getting the inferred position, in about **1.4 seconds**, with an average indoor localization accuracy about **1.3m**.

Literature

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Literature

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Thank U ;-)